



(School of Computer Science and Engineering)

Internship Report
Of
Web Development Using HTML, CSS & JavaScript

Submitted
To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE

By

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Roll Number of Student: CS-2341016

Batch: 23-27

2026-27

SCHOOL OF COMPUTER SCI. AND ENGG.

IILM UNIVERSITY, GREATER NOIDA

CANDIDATE'S DECLARATION

I, Saurabh Shivam, do hereby declare that the internship report titled "Web Development Using HTML, CSS & JavaScript" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and I shall be solely responsible for any kind of copyright violation in this regard.

Signature: _____

Student Name: Saurabh Shivam

Roll No.: CS-2341016

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout this internship. I am thankful for their aspiring guidance, invaluable constructive criticism, and friendly advice during this work. I am sincerely grateful to the team at ApexPlanet Software Pvt. Ltd. for sharing their truthful and illuminating views and supervision throughout the completion of this internship.

I express my gratitude to my parents for their blessings and encouragement.

Signature: _____

Student Name: Saurabh Shivam

Roll No.: CS-2341016

INTERNSHIP COMPLETION CERTIFICATE

CERTIFICATE OF COMPLETION

INTERNSHIP PROGRAM

This is to certify that

SAURABH SHIVAM

has successfully completed the **Internship Program** at **ApexPlanet Software Pvt Ltd** in the domain of **WEB DEVELOPMENT WITH HTML, CSS, AND JAVASCRIPT** from **01 December, 2025** to **31 December, 2025**

Duration: 4 Weeks & 2 Days | Mode: Virtual/Online

During the internship period, the intern actively participated in assigned tasks, project development activities, and practical learning exercises.

We appreciate the dedication and efforts shown throughout the internship and wish them success in future endeavors.



Authorized Signatory
Founder & CEO, ApexPlanet Software Pvt. Ltd.



Certificate ID: APSPL2646249
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4. PROJECT DESCRIPTION

4.1 Introduction

Web development has become one of the most essential skills in the modern technology industry, forming the foundation for building websites, web applications, and digital products used by businesses and individuals worldwide. This internship focused on gaining practical, hands-on experience in front-end web development using HTML, CSS, and JavaScript, followed by exposure to API integration and modern deployment practices.

The internship was structured as a task-based, self-paced program in which each task built upon the skills learned in the previous one — beginning with static webpage structure and styling, progressing to interactivity with JavaScript, then to dynamic data-driven applications using external APIs, and finally to performance optimization, code quality, and deployment. This progression allowed for a well-rounded understanding of how a modern website is designed, built, optimized, and shipped to production.

4.2 Organization Profile

ApexPlanet Software Pvt. Ltd. is a software development company that specializes in delivering web development and app development services to clients while also nurturing new talent through structured internship programs. The organization is dedicated to driving innovation in digital solutions and follows industry best practices to build robust, scalable, and user-friendly applications.

As part of its commitment to developing future tech talent, ApexPlanet Software Pvt. Ltd. runs a comprehensive Web Development Internship Program that provides interns with hands-on project work, mentorship, and a well-defined task pipeline covering HTML, CSS, JavaScript, API integration, performance optimization, and deployment. The internship gave practical exposure to real-world development workflows, version control using GitHub, and professional documentation and submission practices.

During the internship, I worked in the Web Development domain, completing tasks related to building responsive websites, developing interactive JavaScript components, integrating third-party APIs, optimizing site performance and SEO, and finally testing and deploying the completed project.

4.3 Problem Statement

The primary objective addressed during this internship was to design and build a fully responsive, multi-page website from scratch and progressively enhance it with interactivity, live data integration, and performance optimizations, following professional web development standards. Given the growing demand for fast, accessible, and mobile-friendly websites, the internship aimed to develop the practical skills required to plan, build, test, and deploy a production-ready website using core front-end technologies, while maintaining clean, well-organized, and maintainable code.

4.4 Project Objectives

- To design and build a responsive, multi-page static website using semantic HTML5 and CSS3.
- To style layouts using Flexbox, CSS Grid, the Box Model, and media queries for full responsiveness.
- To add interactivity to the website using vanilla JavaScript, including menus, dark/light mode, sliders, modals, and form validation.

- To integrate third-party REST APIs (such as OpenWeatherMap) and build functional mini-applications like a Weather App and a Todo List App.
- To optimize the website for performance, SEO, and accessibility.
- To organize the codebase using modular CSS and JavaScript files and professional naming conventions.
- To test the website across browsers and devices, and deploy it to a live hosting platform.

4.5 Scope of the Project

The scope of this internship project is to build a complete, responsive website that demonstrates the practical application of front-end web development concepts — from static structure and styling to dynamic, API-driven functionality. The project can be extended in the following areas:

- Backend integration: The static/JS front end can be connected to a PHP & MySQL backend for dynamic content management.
- Additional API integrations: More third-party services (maps, payments, authentication) could be added to widen functionality.
- Progressive Web App (PWA): The website can be converted into an installable, offline-capable PWA.
- Advanced frameworks: The vanilla JS implementation can be migrated to a framework such as React for scalability.
- Continuous deployment: CI/CD pipelines could be added for automated testing and deployment.

Overall, the project provides a strong foundation in front-end web development while leaving room for further growth into full-stack development.

4.6 Technologies and Tools Used

- HTML5 — used to build the semantic structure of all website pages, including forms, tables, and layout elements.
- CSS3 — used for styling, including Flexbox, Grid, the Box Model, and media queries to achieve full responsiveness across mobile, tablet, and desktop.
- JavaScript (Vanilla JS) — used to add interactivity such as menu toggles, dark/light mode, image sliders, modals, form validation, smooth scrolling, and animated counters.
- Fetch API / Promises / Async-Await — used for asynchronous data fetching and JSON handling.
- OpenWeatherMap API — a third-party REST API used to build a functional Weather App with live data.
- Browser localStorage — used to persist user preferences and Todo List data between sessions.
- Git & GitHub — used for version control and hosting the project's source code in public repositories.
- VS Code — the primary code editor, used with Live Server and Prettier extensions for an efficient development workflow.
- Google PageSpeed Insights — used to test and improve website performance.
- Netlify / Vercel — used to deploy the finished website to a live, publicly accessible URL.
- LinkedIn — used to document and share progress on each completed task as part of the professional submission process.

4.7 Internship Task Structure

The internship followed a structured, five-task pipeline spread across 30 days, with each task building on the previous one:

- Task 1 — Static Website with HTML & CSS (6 Days): Building a responsive, 4-page static website (Home, About, Services, Contact).
- Task 2 — JavaScript Interactive Components (7 Days): Adding interactive features such as menus, dark mode, sliders, modals, and form validation using vanilla JavaScript.
- Task 3 — API Integration & Mini Applications (7 Days): Building a Weather App using the OpenWeatherMap API and a Todo List App using localStorage.
- Task 4 — Performance, SEO & Code Quality (6 Days): Optimizing images, improving PageSpeed scores, adding SEO meta tags, improving accessibility, and modularizing code.
- Task 5 — Testing, Deployment & Final Submission (4 Days): Cross-browser testing, writing documentation, deploying the website, recording a demo video, and final submission.

4.8 Methodology

Task 1: Static Website with HTML & CSS

The first stage involved setting up the development environment in VS Code with the Live Server and Prettier extensions, and creating a GitHub repository with a proper folder structure. Core HTML5 tags were used to build the structure, forms, tables, and semantic elements for a homepage. CSS was then used to style the navigation bar, hero section, card layouts, and footer using Flexbox, Grid, the Box Model, and media queries to make the site fully responsive across mobile, tablet, and desktop screens. The task concluded with a complete 4-page website — Home, About, Services, and Contact — including a contact form, an embedded Google Maps iframe, and an image gallery, with a consistent header and footer across all pages.

Task 2: JavaScript Interactive Components

This stage focused on JavaScript fundamentals — variables, functions, arrays, objects, DOM manipulation, and event handling (click, submit, scroll, keyup). These fundamentals were applied to build core components such as a mobile hamburger menu toggle, a dark/light mode toggle with preferences saved in localStorage, and a “Back to Top” button that appears after scrolling. Advanced components were then added, including an auto-playing image slider/carousel with manual navigation, a modal popup that can be opened and closed via a button or an outside click, real-time form validation for name, email, and phone fields, smooth scrolling for anchor links, and animated counters that count up when they become visible on screen.

Task 3: API Integration & Mini Applications

The third stage introduced asynchronous JavaScript concepts — the Fetch API, Promises, and async/await — along with JSON parsing and error handling, tested initially using free APIs such as JSONPlaceholder and Random User. These concepts were then applied to build a Weather App using the OpenWeatherMap API, allowing users to search for a city and view its temperature, humidity, wind speed, and weather icon, with proper error handling for invalid city names and the last searched city saved in localStorage. A Todo List App was also built, supporting adding, editing, deleting, and marking tasks as complete, filtering tasks by status (All, Active, Completed), and persisting tasks in localStorage so that they remain after a page refresh.

Task 4: Performance, SEO & Code Quality

This stage focused on optimizing the website and improving code standards. Images were compressed and converted to WebP format, lazy loading was added, and CSS and JavaScript files were minified. The website was tested using Google PageSpeed Insights, achieving a performance score of 88/100. SEO and accessibility were improved by adding meta tags (title, description, keywords, author), Open Graph tags for social media sharing, a favicon, alt text for images, semantic HTML, ARIA labels, and keyboard navigation support. The codebase was then reorganized by splitting CSS into multiple files (header.css, footer.css, home.css) and JavaScript into modules (ui.js, api.js, utils.js), following the BEM naming convention for CSS and adding proper comments while removing unnecessary console logs.

Task 5: Testing, Deployment & Final Submission

The final stage involved cross-browser testing on Chrome, Firefox, Safari, and Edge, as well as testing on mobile devices using Chrome DevTools, and fixing any browser-specific issues found. A professional README.md file was written covering the project overview, features, technology stack, screenshots, and live link, and the website was deployed using Netlify to obtain a live demo URL. A 3-5 minute demo video was recorded showcasing the website tour, dark mode, the Weather App, the Todo App, and the mobile-responsive view, and uploaded to YouTube. The internship concluded with a final, cleaned-up GitHub repository (tagged v1.0.0), a final LinkedIn submission post containing the GitHub link, live demo link, and YouTube link, and submission of all deliverables to ApexPlanet.

4.9 Expected Outcomes

This internship successfully demonstrates the ability to design, build, and deploy a complete, responsive, and interactive website using HTML, CSS, and JavaScript. The completed project integrates static page design, dynamic JavaScript-driven interactivity, real-time API data, and performance/SEO best practices into a single, production-ready deliverable. The structured, task-based approach provided practical exposure to real-world front-end development workflows, version control, and deployment practices, and serves as a strong foundation for further growth into full-stack and framework-based (e.g., React) development in future projects.

4.10 Certificate of Completion

The internship was successfully completed, and a Certificate of Completion (Certificate ID: APSPL2646249) was issued by ApexPlanet Software Pvt. Ltd., confirming successful completion of the Internship Program in the domain of Web Development with HTML, CSS, and JavaScript, for the duration 01 December 2025 to 31 December 2025 (4 Weeks & 2 Days, Virtual/Online mode). The certificate is reproduced below.

CERTIFICATE OF COMPLETION

INTERNSHIP PROGRAM

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- GitHub Documentation: <https://docs.github.com/>
- Netlify Documentation: <https://docs.netlify.com/>
- W3C Web Accessibility Initiative (WAI): <https://www.w3.org/WAI/>